Name:_____ Class:_____ Student number:_____

Why do young shootsgrow towards light?

Clues from everyday life

Why do coleoptiles（胚芽鞘） near a window bend towards light?

This experiment uses oat coleoptiles（胚芽鞘）. A coleoptile（胚芽鞘） is a sheath that covers and protects the young leaves inside. Its fairly straight shape makes changes easy to observe.

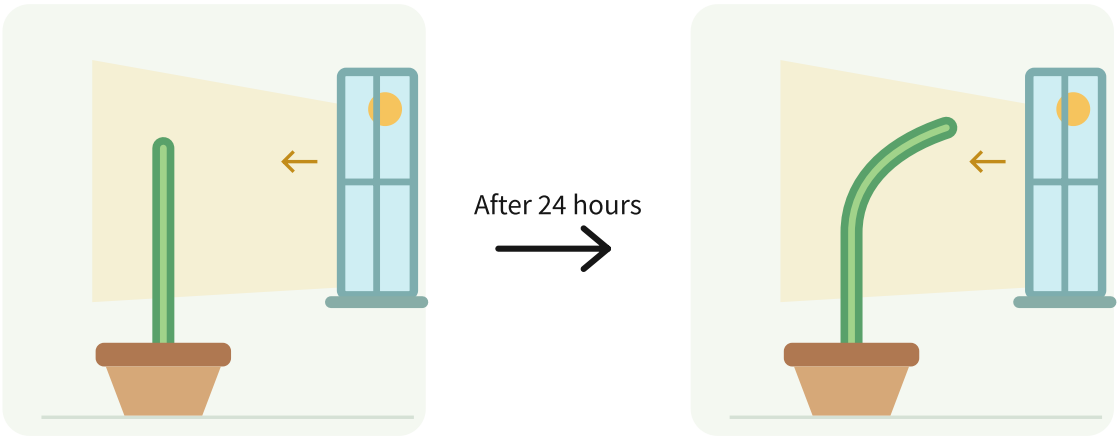


Illustration of growth changes; 24 hours is the time chosen for this situation.

The school gardening club notices that coleoptiles（胚芽鞘） near a window gradually bend towards it. They know that plants grow towards light, but they do not know which part senses light or what makes the plant bend. Can you find clues through experiments?

Your first observations

Describe the changes you can observe. How do the coleoptiles（胚芽鞘）' shape and growth direction change?

(Record observations only. You do not need to explain them here.)

Research question

Which part of a coleoptile（胚芽鞘） senses light from one side and affects its growth direction?

01 • Build a hypothesis

Complete your hypothesis

If the coleoptile（胚芽鞘）'s （Tip ／ Part below the tip） is covered to block light while the other parts still receive light, it will （still bend towards light ／ show no clear bending towards light）.

My reason (required; there is no single answer)

Your full original hypothesis and reason will be saved before your first experiment as part of your inquiry record.

02 • Plan a fair test

Change only the light-blocking treatment. Compare no light blocked, light blocked at the tip and light blocked below the tip. All three setups keep their tips, and other conditions stay the same.

Independent variable(the factor changed on purpose)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Dependent variable(the result measured or observed)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Controlled variables(the factors kept the same)

- ☐ Time exposed to light
- ☐ The coleoptile（胚芽鞘）’s bending response
- ☐ Light direction and intensity
- ☐ Coleoptile（胚芽鞘） type, starting height and growth stage
- ☐ Light-blocking treatment
- ☐ Temperature and water supply

Revision (if needed)

03 • Experiment assumptions

What assumptions does this inquiry make?

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

Choose all suitable assumptions. You may choose more than one.

☐ The coleoptiles（胚芽鞘） start in similar growth conditions.

☐ The light-blocking cap and sleeve do not limit elongation（延長） or bending.

☐ The treatments do not cause clear temperature differences between setups.

☐ If the plants are the same type, their starting size does not need to be controlled.

04 • The control group

To find out whether blocking light affects the coleoptile（胚芽鞘）'s growth direction, how would you design the control group?

(Hint: The control group provides a baseline for comparison. It helps you confirm that a change is caused by the independent variable rather than other external factors, so the control group does not receive the independent variable treatment.)

05 • Experiment setup

Draw and label your experiment setup

Use your experiment variables to draw and label your setup.

(Hint: How many setups do you need? Which are the test groups and the control group?)

Oat coleoptiles（胚芽鞘） × 3; Light source on one side × 3; Opaque（不透光） cap × 1; Light-blocking sleeve for the lower part × 1; Ruler × 1; Timer × 1

Draw, upload a photo or write your plan, then press Save below to save them together.

You do not need to draw every detail carefully. A simple drawing with clear labels is enough.

Written plan (may replace a drawing): describe setups A, B and C, how light is blocked and the conditions kept the same.